

N+ - Network Fundamentals and building network

❖ Wireless Networking

[Q-1] List 3 popular Wi-Fi standards (e.g., 802.11n, 802.11ac, 802.11ax) and for each, write one key difference in speed or frequency band.

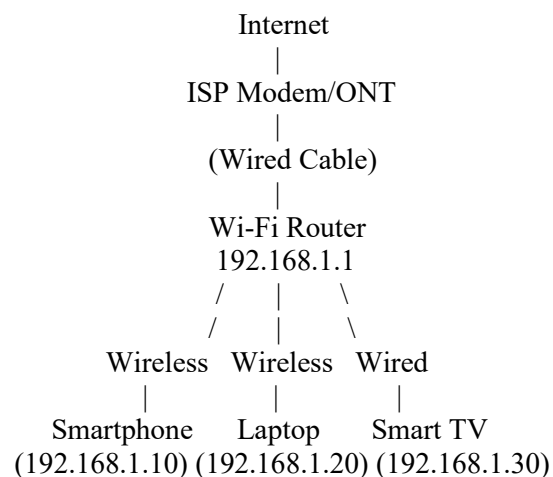
[ANS]

➤ Popular Wi-Fi Standards

Wi-Fi Standard	Frequency Band	Maximum Speed	Key Difference
802.11n (Wi-Fi 4)	2.4 GHz and 5 GHz	Up to 600 Mbps	First standard to support both 2.4 GHz and 5 GHz bands.
802.11ac (Wi-Fi 5)	5 GHz only	Up to 6.9 Gbps	Faster speeds and less interference due to 5 GHz operation.
802.11ax (Wi-Fi 6)	2.4 GHz and 5 GHz (and 6 GHz in Wi-Fi 6E)	Up to 9.6 Gbps	Better performance in crowded networks and higher efficiency.

[Q-2] Draw a simple diagram (on paper or using any drawing tool) showing how your smartphone connects to the internet via a Wi-Fi router at home, labeling the wireless and wired parts of the network.

[ANS]



➤ Labels

- **Wired Connection:** Internet → ISP Modem → Wi-Fi Router
- **Wireless Connection:** Router → Smartphone/Laptop
- **Wired Connection:** Router → Smart TV (Ethernet Cable)

[Q-3] Scan for available Wi-Fi networks using your laptop or phone, note down the SSID, signal strength, and security type (WPA2, WPA3, etc.) for at least 3 networks, and explain which one you would choose and why. Hint: Consider both signal strength and security when choosing.

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SSID	Signal Strength	Security Type
Home WiFi	-42 dBm	WPA3
JioFiber 5G	-58 dBm	WPA2/WPA3
Neighbor WiFi	-75 dBm	WPA2

➤ Which Network Would I Choose?

- I would choose **Home_WiFi** because:
 1. It has the **strongest signal** (-42 dBm).
 2. It uses **WPA3 security**, which is more secure than WPA2.
 3. A stronger signal provides better speed and stability.

➤ Signal Strength Guide

Signal Strength	Quality
-30 to -50 dBm	Excellent
-51 to -67 dBm	Good
-68 to -80 dBm	Fair
Below -80 dBm	Weak

[Q-4] Explain in your own words what 'SSID hiding' is in Wi-Fi networks and discuss one advantage and one disadvantage of hiding your Wi-Fi SSID.

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➤ What is SSID Hiding?

- SSID (Service Set Identifier) is the name of a Wi-Fi network. When SSID hiding is enabled, the router does not broadcast the network name publicly. Users must manually enter the Wi-Fi name to connect.

➤ Normal Example

- Available Networks:
 1. Home_WiFi
 2. JioFiber_5G
 3. Office_Network

➤ SSID Example

- Available Networks:
 1. Hidden Network
 2. JioFiber_5G
 3. Office_Network

➤ Advantage of SSID Hiding

- Casual users cannot easily see the Wi-Fi network in the list of available networks, providing a small layer of privacy.

➤ Disadvantage of SSID Hiding

- It does **not provide real security** because attackers can still detect the hidden SSID using Wi-Fi analysis tools.
- Users must manually enter the network name when connecting new devices.